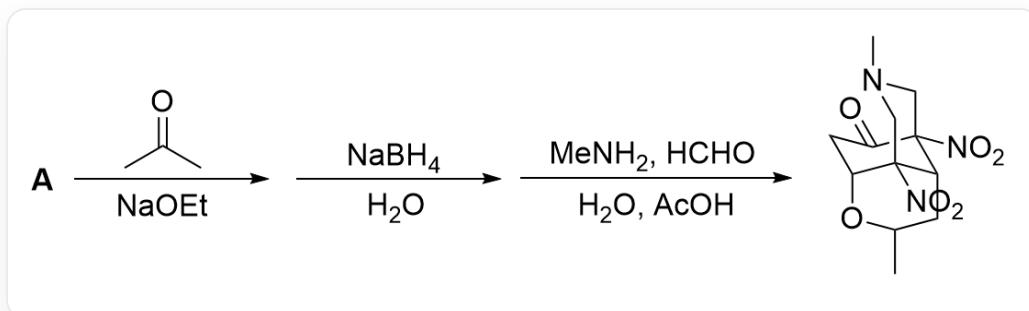


Question

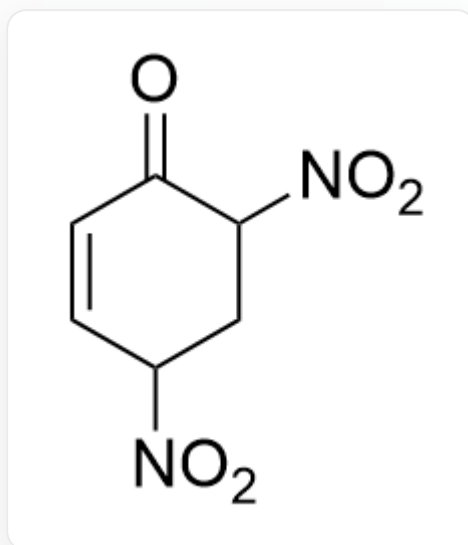


The image describes a one-pot organic cascade reaction. Substrate **A** first reacts with CC(C)=O , NaOEt ; then reacts with NaBH_4 , H_2O ; and finally reacts with MeNH_2 , HCHO , H_2O , AcOH to obtain the product CN1C[C@@]([C@@H]2[C@]3([N+](O-)=O)C1)([N+](O-)=O)[C@H](OC(C)C2)CC3=O

Which of the following options is correct regarding the structure of the unknown structure **A** in the above reaction:

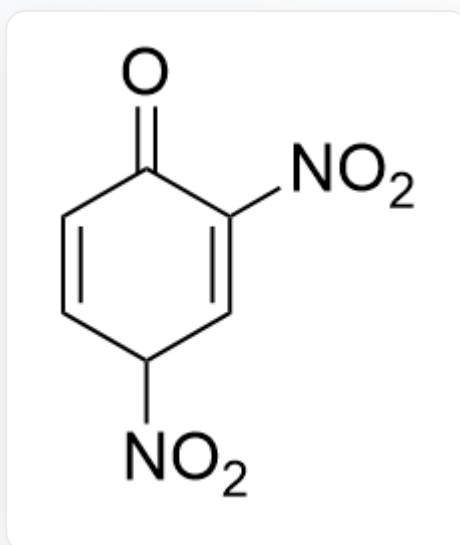
A. All other options are incorrect

B.



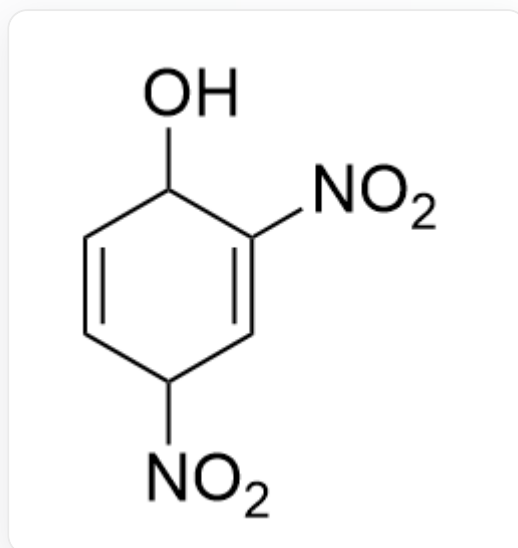
O=C1C([N+](O-)=O)CC([N+](O-)=O)C=C1

C.



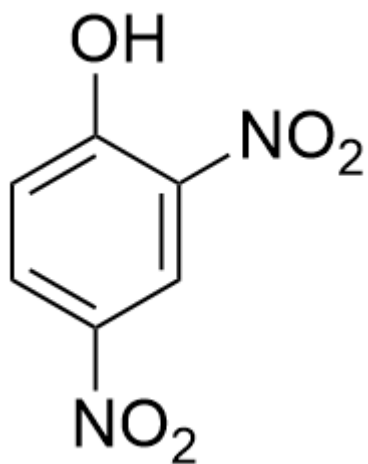
O=C1C([N+](=[O-])=O)=CC([N+](=[O-])=O)C=C1

D.



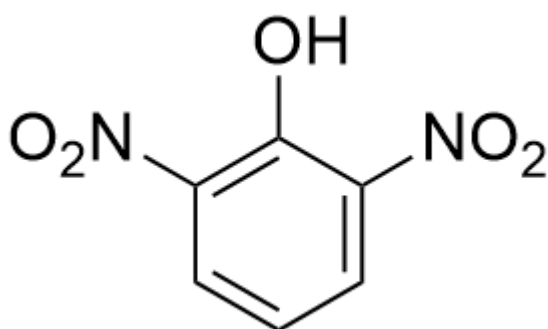
OC1C([N+](=[O-])=O)=CC([N+](=[O-])=O)C=C1

E.



OC1=CC=C([N+](O-)=O)C=C1[N+](O-)=O

F.



OC1=C([N+](O-)=O)C=CC=C1[N+](O-)=O

Answer

Correct Answer: **E**

Detailed Explanation

The last step in the formation of the product is obviously a Mannich reaction; the tertiary amine unit $-\text{CH}_2 - \text{NCH}_3 - \text{CH}_2-$ in the product is the structure obtained from the Mannich reaction, so the first step of retrosynthesis requires cleaving this tertiary amine fragment to generate a dicarbanion, O=C1[C-]([N+])([O-])=O [C@H]2[C-]([N+])([O-])=O)[C@H](OC(C)C2)C1.

CHECKPOINT

1 PTS

The last step in the formation of the product is a Mannich reaction

CHECKPOINT

1 PTS

The tertiary amine unit $-\text{CH}_2 - \text{NCH}_3 - \text{CH}_2-$ in the product is the structure obtained from the Mannich reaction

CHECKPOINT

1 PTS

Cleave this tertiary amine fragment to generate a dicarbanion with the structure O=C1[C-]([N+])([O-])=O [C@H]2[C-]([N+])([O-])=O)[C@H](OC(C)C2)C1

In fact, the negative charge of the carbanion can be transferred to the nitro group, i.e., the $C = NO_2^-$ double bond structure. Therefore, the structure cleaved in the first step is O=C(C[C@@H]1/C([C@H]/2CC(C)O1)=[N+])([O-])[O-]C2=[N+](O-)[O-].

CHECKPOINT

1 PTS

The structure cleaved in the first step is O=C(C[C@@H]1/C([C@H]/2CC(C)O1)=[N+])([O-])[O-]C2=[N+](O-)[O-]

In the next step, sodium borohydride was added. Since it is obvious that the $-O-CHCH_3-CH_2-$ six-membered ring fragment of the product comes from the initially added acetone, consider that sodium borohydride is used to reduce the carbonyl group; however, because the product contains a carbonyl group, the carbonyl group of the product can only be reduced when it is an α,β -unsaturated ketone structure to prevent it from being reduced. Therefore, after the carbonyl group of the acetone structure is reduced to an alcohol hydroxyl group, an intramolecular Michael addition occurs, and the nucleophilic attack on the α,β -unsaturated ketone forms a six-membered ring, which is logical. Therefore, the structure cleaved further is O=C(C=C/C(C/1CC(C)=O)=[N+](O-)[O-])C1=[N+](O-)[O-].

CHECKPOINT

1 PTS

The $-O-CHCH_3-CH_2-$ six-membered ring fragment of the product comes from acetone

CHECKPOINT

1 PTS

Sodium borohydride is used to reduce the carbonyl group

CHECKPOINT

1 PTS

After the carbonyl group of acetone is reduced to an alcohol hydroxyl group, an intramolecular Michael addition occurs, and a nucleophilic attack on the α,β -unsaturated ketone forms a six-membered ring

CHECKPOINT

1 PTS

The structure cleaved further is O=C(C=C/C(C/1CC(C)=O)=[N+][O-])C1=[N+][O-]

The combination of the acetone structure is obviously achieved by converting acetone into an enolate under sodium alkoxide conditions for nucleophilic attack, so cleaving the acetone structure yields a six-membered ring carbocation, with the structure O=C(C=C/C([CH+]/1)=[N+][O-])C1=[N+][O-].

CHECKPOINT

1 PTS

Cleaving the acetone structure yields a six-membered ring carbocation, with the structure O=C(C=C/C([CH+]/1)=[N+][O-])C1=[N+][O-]

This structure now carries a negative charge and needs to obtain a hydrogen ion; it is finally found that **A** contains four hydrogens and can tautomerize to a benzene ring structure, while the nucleophilic attack of the anion generated by acetone is an aromatic nucleophilic substitution reaction.

CHECKPOINT

1 PTS

A contains four hydrogens and can tautomerize to a benzene ring structure

CHECKPOINT

1 PTS

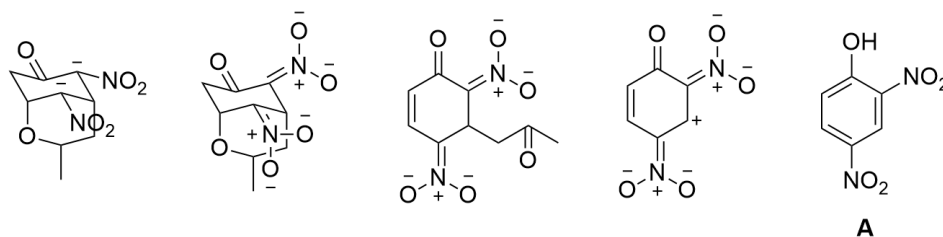
The nucleophilic attack of the anion generated by acetone is an aromatic nucleophilic substitution reaction

Therefore, the structure of **A** is OC1=CC=C([N+][O-])C=C1[N+][O-], and option E is correct.

CHECKPOINT

2 PTS

The structure of **A** is OC1=CC=C([N+][O-])C=C1[N+][O-]



This figure shows the intermediates of the product being cleaved step by step from left to right. Cleavage of the tertiary amine fragment generates a dicarbanion, with the structure O=C1[C-]([N+][O-])C@H]2[C-]([N+][O-])=O)[C@H](OC(C)C2)C1; tautomerization, to O=C(C[C@@H]1/C([C@H]2CC(C)O1)=[N+][O-])[O-])C2=[N+][O-][O-]; then cleavage of the six-membered ring, with the structure O=C(C=C/C(C1CC(C)=O)=[N+][O-])[O-])C1=[N+][O-][O-]; cleavage of the acetone structure yields a six-membered ring carbocation, with the structure O=C(C=C/C([CH+]/1)=[N+][O-][O-])C1=[N+][O-][O-]; finally, tautomerization yields the structure of **A** as OC1=CC=C([N+][O-])C=C1[N+][O-].